

PROJECT PROPOSAL

AI-Powered Visual Search for Zen Cart Locomotive Inventory System

Prepared for	Piotr Train Collection Website	Prepared by	Datix AI
Date	August 3, 2026	Est. Timeline	10-14 days

1. Project Overview

Your Zen Cart website currently serves as a private inventory and preparation-request system for your train collection not an online store. Visitors browse inventory, add locomotives and rolling stock to a cart, and submit the list to you by email for review and preparation. Inventory lookup is currently handled through QR code stickers placed under each locomotive, which link directly to the correct product page.

This proposal covers adding a Visual Search feature: a customer or you point a phone camera at a locomotive, and the system attempts to identify it and open the matching product page automatically as an additional lookup method alongside the existing QR code system (which will remain in place as the reliable fallback).

2. Scope of Work

What's included

- Review of the existing Zen Cart codebase and database structure (products table, SKUs, image paths).
- Setup of a cloud-based AI image recognition project (Azure Custom Vision or Google Vertex AI — final choice confirmed after reviewing your reshoot photos).
- Training of a custom image-classification model using your locomotive photos, tagged by SKU/product ID.
- A new camera-upload page built into your existing Zen Cart site (no site rebuild, no platform migration).
- Integration logic: uploaded photo → sent to AI service → best-match product ID(s) returned → redirect to the matching product page.
- A fallback "Top 3 matches" screen for cases where the model isn't confident enough for a single automatic redirect expected for very similar locomotives.
- Testing using real phone photos (not just clean training images) and threshold tuning.
- A short handover note explaining how to add new locomotives to the visual search index in future.

What's not included

- Migrating the website to WordPress/WooCommerce or any other platform.
- Re-enabling credit card processing or e-commerce checkout functionality.
- Guaranteed identification accuracy for near-identical locomotives (see Limitations, Section 6).

- Ongoing AI service usage fees beyond setup (see Section 4).

3. Approach & Timeline

1. Access & discovery (Day 1–2): Review cPanel/staging access, database structure, and existing product photo set.
2. Photo dataset preparation (parallel, on your side): You reshoot/organize locomotive photos solid background, good lighting, multiple angles per item — while setup work proceeds.
3. Model setup & training (Day 2–5): Upload tagged images, train initial model, review per-item accuracy results.
4. Model refinement (Day 5–7): Add extra images for locomotives the model confuses with each other; retrain.
5. Zen Cart integration (Day 6–9): Build the camera-upload page and connect it to the AI service and your product database.
6. Testing & tuning (Day 9–12): Real-world phone photo testing, confidence threshold tuning, fallback screen refinement.
7. Review & handover (Day 12–14): Walkthrough with you, adjustments based on your feedback, documentation handover.

Total estimated timeline: 10–14 days from receiving your reshot photos and site access, depending on collection size and how many locomotives need extra training images.

4. Investment

Item	Cost (USD)
Development fee — visual search build, per scope above	Not Confirmed yet
Cloud AI service (Azure Custom Vision / Vertex AI) — model training & hosting	Billed separately, your own account
Ongoing per-scan API cost after launch	Usage-based, typically small confirmed once provider is chosen

The final fee within the \$250–\$350 range depends on collection size and how much refinement the model needs. It will be confirmed once your photo set and site access are reviewed. This is a flat project fee, not hourly.

Recommended: the cloud AI account (Azure/Google) is created under your own billing, so ongoing usage costs are visible to you directly and there's no markup or pass-through.

5. What I Need From You

- cPanel/FTP access, or a staging clone of the site (your choice — staging is safer for testing new pages).
- Database access details (or confirmation I can pull them from Zen Cart's configure.php once inside cPanel).
- Reshoot photos of each locomotive: solid/white background, good even lighting, at least 2–3 angles per item where possible.
- Confirmation of which locomotives are most visually similar to each other, if you already know — this helps prioritize where extra training images matter most.
- A decision on Azure vs. Google for the AI service, or approval for me to recommend one after reviewing the photo set.

6. Limitations & Expectations

- Visual search will not be perfectly accurate for locomotives that are nearly identical (same manufacturer, era, and paint scheme) this is a known limitation of image recognition, not a defect in the build.
- For low-confidence matches, the system will show a short list of likely candidates rather than guessing a single wrong answer.
- Accuracy depends heavily on photo quality — poorly lit or cluttered training photos will reduce reliability.
- The QR code system remains the most reliable lookup method and will not be removed; visual search is an additional, best-effort option.
- A small proof-of-concept (10–15 locomotives) can be run first if you'd like to see real accuracy numbers before committing photos for the full collection.
- This proposal assumes no major structural issues in the existing Zen Cart codebase; anything unexpected found during discovery will be flagged before extra work begins.

7. A Few Questions Before I Begin

8. Do you want to run a small proof-of-concept first (10–15 locomotives) before committing photos for the whole collection, or go straight to the full build?
9. Would you prefer I work on a staging copy of the site, or directly on the live cPanel environment?
10. Roughly how many locomotives/items are in the current inventory that would need training photos?
11. Are you comfortable creating your own Azure or Google Cloud billing account for the AI service, or would you like a recommendation on which to use first?
12. Is there a target date you're hoping to have this live by (e.g. a show, event, or season), or is 10–14 days from photo delivery flexible?

Looking forward to your confirmation.



ARTIFICIAL INTELLIGENCE & DATA SCIENCE

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